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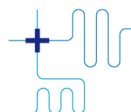
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Palm Fruit Harvesting Using IoT-Based Fruit Counting System

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Abstract. This research focuses on palm fruit harvesting using data from Internet of Things (IoT) technologies. Alternatives to relying on error, one can might consider using sensors for tracking. However, advanced sensors can be expensive. Therefore, we are considering cheaper options with simplified sensors. One form detects movement, while the other detects pressure. These choices could simplify fruit counting without spending much. Our research involves past studies, understanding parts, and how they connect.

INTRODUCTION

For palm oil ripeness detection system, fruit counting system is considered as one of the important parts of this system. To ensure the harvesting schedule and for growers to maximize their production and make choices regarding when to harvest their crops, it is crucial to estimate fruit count accurately [1]. Robotic fruit counting systems use many components such as cameras, depth sensors, global positioning system (GPS) [2]. Although this method has been effective for counting fruits such as oranges or mangoes, this system requires super high cost and weight [3]. Hence, an idea has been suggested and is being used for this fruit counting system [4,5]. Which type of sensors and how the sensors are used for this system will be explained in this chapter.

LITERATURE REVIEW

Based on a paper named IoT Enabled Intelligent Sensor Node for Smart City: Pedestrian Counting and Ambient Monitoring researched by [6], PIR sensor is used as human counter for pedestrian counting. A total of three pairs of PIR sensors which are connected to the analogue inputs of Arduino Uno are used for counting pedestrians [7]. An algorithm has been developed to accurately count the number of pedestrians within the field of view of the PIR sensor, allowing the use of the PIR sensors as a human counter. This algorithm makes use of a flag variable and a counter variable [8]. Firstly, the flag variable will be set at a LOW state and the output will be LOW for the PIR sensor. Once someone enters the focus zone of PIR sensor, then it will change from LOW to HIGH state. The timer can be set to any value, and it is designed so that it takes roughly the same amount of time for a human to walk the length of its focus angle no matter what that angle is. Hence, whenever the sensor changes from LOW to HIGH, it will be counted incrementally. Lastly, as soon as the pedestrian crosses the focus zone, it will then set back to LOW and wait for the next pedestrian to pass through.

Based on the study made by [9] which named Design and Accuracy of an Instrumented Insole using Pressure Sensors for Step Count. The sensor used is Force Sensitive Resistor (FSR). FSR is a passive component that exhibits a decrease in resistance as the force applied to the sensing region increases [10]. The pressure sensors were embedded in a commercial insole that was designed to be adaptable, comfortable, lightweight, and inexpensive. Whenever there is a step, there will be a change in resistor on the FSR, thus the microcontroller will process the signal [11]. There will be an algorithm for the step count. Step count algorithm involved increasing the step number for each pressure peak that matched to the period when the fruit contacted the ground. Pressure data from each FSR separately and from all FSRs combined were used to build the step count algorithms. There will be 5 FSR in total, whenever a person takes a step, there will be different pressure applied on the FSR, hence by calculating the pressure applied, step counting can be easily recorded.

Researcher [12] stated that loose fruit has the most oil content compared to other fruits in bunch. Loose fruits can be categorized into two classes, one to those detached when fresh fruit bunch (FFB) is ripe and second those that are scattered on ground [13]. Depending on the variety and age, the weight of FFB can range between 10 to 30kg while one fruit is around 6 to 20g [14]. Loose fruits will normally drop and scattered in a radius of 1.5m to 2.5m from the fresh fruit bunch (FFB) point. According to the paper, it stated that it is a common practice that when there are 1 to 10 loose fruits dropped on the ground, it means that the harvesting can be performed as soon as possible as the bunch is already ripe enough [15].

INVESTIGATION ON MATERIALS AND COMPONENT SELECTION

Since this is a fruit counting system, motion sensor and pressure sensor will be important to act as a fruit counting system [16, 17]. Based on studies, pressure sensor can be used to calculate step count and motion (PIR) sensor can be used to detect pedestrian cross, therefore by using the concept of step counting and pedestrian crossing, pressure sensor and motion sensor have been selected as the components to run this fruit counting system. Motion sensor will be used to detect the falling of loose fruit while for pressure sensor, it will be used to detect the weight of the loose fruit and acts as a fruit counting system. Pressure sensors and motion sensors will be connected to Arduino UNO, but pressure sensors will relate to a resistor while motion will not. Sensor fusion with AI can improve the overall data obtained [18-20].

METHODOLOGY

When system start, firstly, N will be set to 0, N is basically the amount of fruit drop, since no fruit is dropping, hence it is set to be 0. Next, the system will start to get signal from motion sensor. As soon as motion sensor detects a fruit drop, the system will be turn on from sleep mode.

Next, it will then start to get signal from pressure sensor or motion sensor, and as soon as pressure sensor or motion sensor detects a fruit drop, it will then send a signal to colour sensor, once colour sensor detects that the fruit is ripe then it will count inside the fruit count N will become 1 and this process will be continue by having the equation of $N=N+1$, once the fruit count reaches 5, it will then send data to cloud telling owner that the palm tree is ready to harvest. If the colour sensor detected that the fruit drop is not ripe, it will not count inside the $N=N+1$ equation but will continue to get input from motion sensor.

This fruit counting system basically consists of a motion sensor and a pressure sensor. Once motion sensor or pressure sensor detected that there is a fruit dropping it will then send signal to Arduino Mega. Next the signal will then being send to ripeness detection system to ensure that the fruit is ripe and to IoT Cloud system if the drop loose fruit reaches 5 for harvesting purposes.

CONCEPT DESIGNED DERIVED FROM FUNDAMENTAL ENGINEERING PRINCIPLE

Figure 1 shows the connection circuit for the fruit counting system. This fruit counting system consists of one motion sensor and one pressure sensor. PIR motion sensor uses digital pins while for pressure sensor, it uses analogue pin. Hence, PIR motion sensor is being connected to ground, voltage, and digital pin 6 while for pressure sensor, one 1k Ω resistor is being used and connected to analogue pin A0 and ground while the other is being connected to voltage.

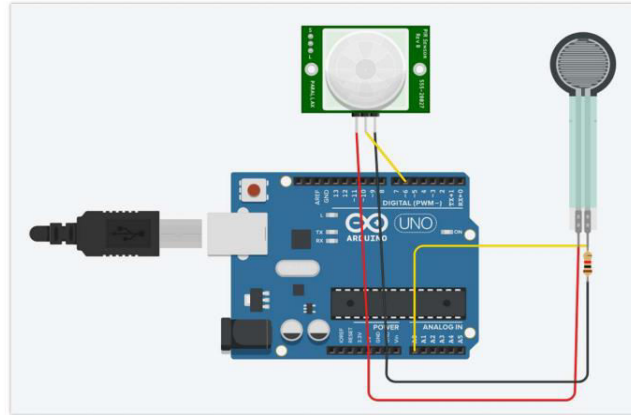


FIGURE 1. Circuit Connection for Fruit Counting System

SYSTEM IMPLEMENTATION

The Arduino Microcontroller used is Arduino Mega. Firstly, is basically integer all the used pins and terms such as nonfruit, pressure, count and many others. Next, under void setup, it includes serial begin and showing the pin mode. Next, for void loop, first is basically reading the analogue pressure pin and serial monitor will print the pressure pad reading. If pressure pad reading is lesser than 0 or motion sensor is LOW, then serial monitor will print no fruit drop else if pressure pad reading is more than 0 or motion sensor is HIGH, serial monitor will print one fruit drop. Next, $\text{count}=\text{n}++$ means the $N=N+1$, N stands for the fruit count, the fruit count will be increasing based on the fruit drop. Serial monitor will show total fruit count = the number of fruit drop. When the fruit count N is more than 5, then serial monitor will print ready to harvest which means it will inform IoT Cloud system that the tree is ready to be harvested.

RESULTS AND DISCUSSION

The fruit counting system that was constructed for purposes such as detecting palm oil ripeness has been demonstrated to be a crucial element in guaranteeing precise and quick harvesting choices for palm growers. The system used a combination of motion sensors and pressure sensors to accurately quantify and measure the stage of development of unattached palm fruits. The following four points are the principal findings:

- **Sensor Functionality:** Motion sensors (PIR) detected fruit falls, while pressure sensors (FSR) measured fruit weight. This helped the machine spot falling fruits and properly count ripe fruits.
- **Data Processing:** The system continuously monitored sensors and updated fruit counts. The serial monitor showed the total fruit count (N) for monitoring and decision-making.
- **Combine with Ripeness Detection:** The ripeness detecting system and fruit counting system worked together to count only ripe fruits. The accuracy of ripeness assessments improved.
- **System Reliability:** The system consistently counted falling fruits and distinguished ripe from unripe ones.

Figure 2 shows the serial monitor when no fruits is being detected by the pressure or motion sensor. Next if pressure or motion sensor detected one fruit dropping, then serial monitor will print the pressure pad reading and showing one fruit drop. The total fruit count will be 1 since the fruit drop is 1. For second fruit drop to fourth fruit drop, the result will be the same, it's just that the total fruit count will be increasing from 1 to 4. Eventually, for the 5th fruit drop. When it is being detected, one fruit drop will be printed on serial monitor, the total fruit count will also become 5. Lastly, "Ready to harvest" will be printed since the fruit drop already reaches 5 which will be informing IoT and Cloud part that the tree is ready to harvest. Reviewing the interdependence of the parameters could create an opportunity for optimization [20].

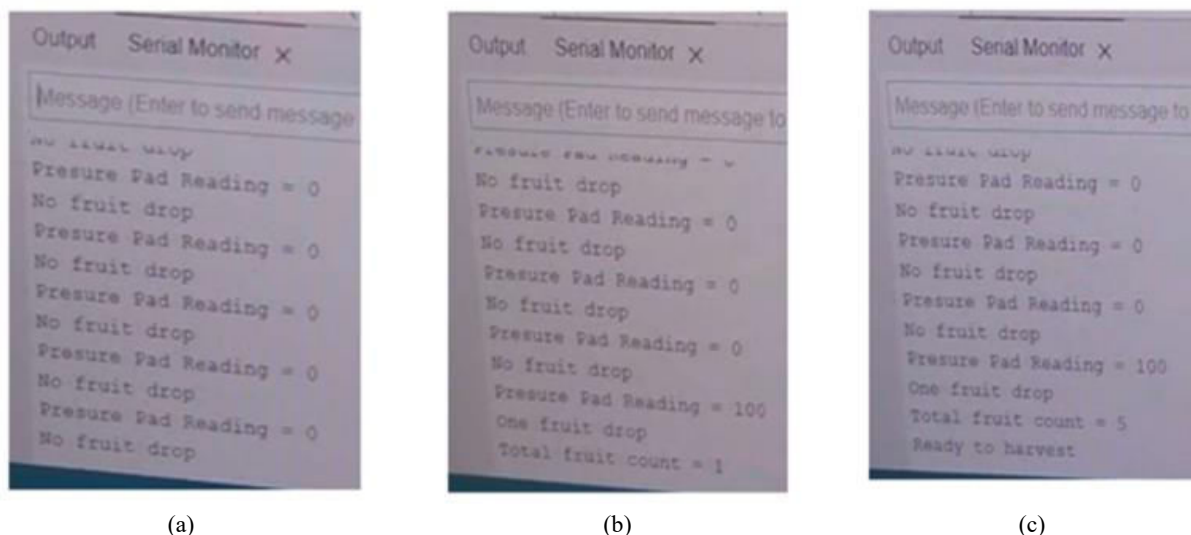


FIGURE 2. Serial Monitor detection: (a) Serial Monitor when no fruit is detected, (b) Serial Monitor when one fruit drops Serial Monitor when five fruits drop, and (c) Serial Monitor when one fruit drops.

CONCLUSION

This fruit counting system is very important to this system because once detected fruit dropping, it will send signal to colour sensor which acts as ripeness detection system. If it detected that the fruit is ripe, then only the fruit will be counted into the fruit count, else it will get back to motion or pressure sensor to detect the fruit drop and once the fruit count reaches 5, it will then send signal to IoT and Cloud system to inform them that the fruit is ready to harvest. Therefore, an Arduino board, a motion sensor, a pressure sensor, jumper wires and a resistor are used to build this fruit counting system.

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